



## 2 Real Analysis 2023

### Tutorial 2

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## Suprema, infima, maxima, minima

**Problem 4** Let  $A \subseteq \mathbb{R}$  be a nonempty set.

- (a) Assume that  $\max A$  exists. Show that  $\max A = \sup A$ .
- (b) Assume that  $A$  is bounded above, and  $\sup A \in A$ . Show that  $\max A$  exists and  $\max A = \sup A$ .
- (c) We denote  $-A = \{-a : a \in A\}$ . Assume that  $A$  is bounded above. Show that  $-A$  is bounded below and

$$\inf(-A) = -\sup A.$$

- (d) Assume that  $A$  is bounded below with  $\inf A > 0$ . We denote  $A^{-1} = \{a^{-1} : a \in A\}$ . Show that  $A^{-1}$  is bounded above and

$$\sup(A^{-1}) = (\inf A)^{-1}.$$

**Problem 5** Determine, without proof, suprema, infima, maxima and minima, if they exist, of the following sets:

- (a)  $A = \left\{ \frac{3n}{2n+1} : n \in \mathbb{N} \right\}$ ,
- (b)  $B = \{[1 + (-1)^n]n : n \in \mathbb{N}\}$ ,
- (c)  $C = \left\{ (-1)^n + \frac{1}{2^n} : n \in \mathbb{N} \right\}$ ,
- (d)  $D = \left\{ \frac{2n}{n+m} : n, m \in \mathbb{N} \right\}$ .

## The Archimedean property and the nested interval property

**Problem 6** (a) Prove that  $\bigcap_{n=1}^{\infty} [n, \infty) = \emptyset$ .

(b) Prove that  $\bigcap_{n=1}^{\infty} \left(0, \frac{1}{n}\right) = \emptyset$ .

(c) Determine  $\bigcap_{n=1}^{\infty} \left[0, \frac{1}{n}\right]$ .